

Math 306 Exercise Set 5

Textbook exercises from Section 5.A: 6, 12, 13, 14, 21

Textbook exercises from Section 5.B: 1, 11

Textbook exercises from Section 5.C: 2, 6, 8

1. The dot product of the vectors $\begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$ and $\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$ in $\mathbb{R}^n$ is defined to be $v_1u_1 + \cdots + v_nu_n$. For a fixed $v \in \mathbb{R}^n$, define $d_v : \mathbb{R}^n \rightarrow \mathbb{R}$ by $d_v(u) = v \cdot u$ for all $u \in \mathbb{R}^n$.

a. Show that d_v is a linear map.

b. Show that the function $T : \mathbb{R}^n \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R})$ defined by $Tv = d_v$ is an isomorphism (an invertible linear map).

2. Suppose that in the standard basis $M(T)$ is the $n \times n$ matrix with diagonal entries 0 and all other entries 1. Find the eigenvalues and eigenvectors for T .